# MSE and MCPE for Regionally Benchmarked MFH Estimates

***Parametric bootstrap procedure implemented in bench_regional_mfh() — revised***

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## 1. Purpose

This note documents how the EU SAE package estimates the mean squared error (MSE) and the mean cross-product error (MCPE) of *benchmarked* small-area poverty rates after fitting a Multivariate Fay–Herriot (MFH) model. The implementation is `scripts/bench_regional_mfh.R`, function `bench_regional_mfh()`, invoked with `MSE = TRUE`. The function supports MFH1, MFH2, and MFH3 through the `model_type` argument; the exposition below uses MFH2 as the running case and notes how MFH1 and MFH3 differ. This revision updates the original note to match the current code, which now incorporates the review comments summarized in Section 9.

This routine is the multivariate, multi-period generalization of the univariate `bench_regional()` documented in the companion UFH note. The two share the same region-level ratio adjustment (Section 3) and the same definition of bootstrap MSE relative to the true small-area value (Section 4); what the univariate case lacks is the time dimension, and with it the AR(1) random-effect structure and the mean cross-product error (MCPE) of Section 5. The model EBLUP is written $\hat{\theta}_{dt}^{\text{EBLUP}}$ here and $\hat{\theta}_{d}^{\text{FH}}$ in that note (the `FH` column of the univariate output); the domain index $d$ and the region index $r$ are common to both.

## 2. Notation

Let $D$ be the number of small areas (domains), indexed $d=1,\ldots,D$; $T$ the number of time periods, indexed $t=1,\ldots,T$; and $R$ the number of regions used as benchmarking constraints, indexed $r$. Write $N_{dt}$ for the population size of domain $d$ at time $t$ — by default the routine uses a single $N_d$ for all periods, but a time-varying $N_{dt}$ may now be supplied (Section 9, point 2). Let $\hat{\theta}_{dt}^{\text{dir}}$ be the direct estimate of the poverty rate in domain $d$ at time $t$, $\hat{\theta}_{dt}^{\text{EBLUP}}$ the MFH EBLUP, and $\theta_{dt}$ the unobserved true area mean.

## 3. Point-estimate benchmarking

Within each region $r$ and period $t$ the routine forms a population-weighted regional benchmark from the direct estimates and a population-weighted regional mean of the model EBLUPs, **on a common set of domains**. Let $B_{rt}$ denote the domains in region $r$ at time $t$ with finite direct estimate, finite EBLUP, and finite population weight. Then

$$B_{rt} = \frac{\sum_{d \in B_{rt}} N_{dt} \hat{\theta}_{dt}^{\text{dir}}}{\sum_{d \in B_{rt}} N_{dt}}, \acute{\hat{\theta}}_{rt}^{\text{EBLUP}} = \frac{\sum_{d \in B_{rt}} N_{dt} \hat{\theta}_{dt}^{\text{EBLUP}}}{\sum_{d \in B_{rt}} N_{dt}}.$$

Using the *same* domain set $B_{rt}$ for both quantities keeps the benchmark and the model-side mean on a consistent weighting basis. Alternatively, an externally estimated benchmark $B_{rt}$ may be supplied directly (Section 9, point 1), in which case $\acute{\hat{\theta}}_{rt}^{\text{EBLUP}}$ is taken over the domains with finite EBLUP and weight. The ratio adjustment factor for region $r$ at time $t$ is

$$\lambda_{rt} = \frac{B_{rt}}{\acute{\hat{\theta}}_{rt}^{\text{EBLUP}}},$$

set to $1$ when the denominator is below $\text{eps\_denom} = 10^{-8}$ or when the benchmark is missing. The benchmarked EBLUP is $\hat{\theta}_{dt}^{\text{bench}} = \lambda_{rt} \hat{\theta}_{dt}^{\text{EBLUP}}$ for $d \in r$. By construction the population-weighted mean of $\hat{\theta}_{dt}^{\text{bench}}$ within region $r$ equals $B_{rt}$.

## 4. MSE of the benchmarked rates: parametric bootstrap

Following Datta, Ghosh, Steorts, and Maples (2011), the MSE of a benchmarked predictor is defined relative to the underlying true small-area mean, not relative to the regional benchmark. For each $(d, t)$ the routine targets

$$MSE\left(\hat{\theta}_{dt}^{\text{bench}}\right) = E\left[\left(\hat{\theta}_{dt}^{\text{bench}} - \theta_{dt}\right)^2\right].$$

Because $\hat{\theta}^{\text{bench}}$ is a non-linear function of the EBLUPs (through $\lambda_{rt}$), there is no closed form, and a parametric bootstrap with $n_B$ replicates (default $n_B = 200$) is used.

### *4.1 Inputs harvested from the fitted MFH object*

Before the loop, the routine extracts the time-$t$ regression coefficients $\hat{\beta}_t$ (model_obj$fit$estcoef); the random-effect variance model_obj$fit$refvar — a scalar $\hat{\sigma}_u^2$ under MFH2, a length-$T$ vector $\hat{\sigma}_{u,t}^2$ under MFH1 and MFH3; the AR(1) coefficient $\hat{\rho}$ (model_obj$fit$rho, used by MFH2 and MFH3); and the $T \times T$ sampling-error covariance matrix $\Sigma_{e,d}$ for each domain, assembled from the vardir columns and forced to positive definiteness via Matrix::nearPD() when necessary.

### *4.2 Data-generating process for replicate b*

For each replicate $b$ the routine simulates from the fitted model. The random effects are generated according to the variant:

For **MFH2** (homoskedastic AR(1)), draw $a_{dt} \sim N\left(0, \hat{\sigma}_u^2\right)$ and use the stationary initialization $u_{d,1}^{(b)} = \left(1 - \hat{\rho}^2\right)^{-1/2} a_{d,1}$, then $u_{dt}^{(b)} = \hat{\rho}\, u_{d,t-1}^{(b)} + a_{dt}$ for $t \geq 2$. For **MFH3** (heteroskedastic AR(1)), the innovations have time-varying variance $a_{dt} \sim N(0, \hat{\sigma}^2_{u,t})$ and the first period is initialized directly as $u^{(b)}_{d,1} = a_{d,1}$ — a stationary distribution does not strictly exist in the heteroskedastic case, so no stationarity factor is imposed (Section 9, point 4). For **MFH1** (independent effects), $u^{(b)}_{dt} \sim N(0, \hat{\sigma}^2_{u,t})$ independently across $t$.

Sampling errors are drawn per domain as $e^{(b)}_{d} \sim N_T(0, \Sigma_{e,d})$. The synthetic truth and observation are

$$\theta^{(b)}_{dt} = x^\top_{dt} \hat{\beta}_t + u^{(b)}_{dt}, \; y^{(b)}_{dt} = \theta^{(b)}_{dt} + e^{(b)}_{dt}.$$

***4.3 Refit, re-benchmark, accumulate***

Each replicate then refits the chosen MFH variant (`eblupMFH1`, `eblupMFH2`, or `eblupMFH3`) on the synthetic data; **re-benchmarks** the bootstrap EBLUPs using the same regional ratio adjustment, with $y^{(b)}_{dt}$ playing the role of the direct estimate, yielding $\lambda^{(b)}_{rt}$ and $\hat{\theta}^{\text{bench},(b)}_{dt}$; and scores the result against the synthetic truth, $\delta^{(b)}_{dt} = \hat{\theta}^{\text{bench},(b)}_{dt} - \theta^{(b)}_{dt}$. Re-running the benchmarking inside each replicate is what propagates the variability of $\lambda_{rt}$ into the MSE.

A replicate whose refit fails to converge is skipped, counted in `fails`, and retried; the loop continues until $n_B$ successful replicates are obtained or a hard cap `max_attempts` ($= 5 n_B$) is reached. The accumulated squared residuals are divided by the number of **successful** replicates $n_{\text{valid}}$ — not by $n_B$ — so convergence failures do not deflate the estimates (Section 9, point 3):

$$\widehat{MSE}\left(\hat{\theta}^{\text{bench}}_{dt}\right) = \frac{1}{n_{\text{valid}}} \sum_{b=1}^{n_{\text{valid}}} \left(\delta^{(b)}_{dt}\right)^2.$$

## 5. MCPE of the benchmarked rates

The MCPE captures the across-time covariance of the prediction error within a domain. For each pair $(t_1, t_2)$ with $t_1 < t_2$,

$$\widehat{MCPE}_d(t_1, t_2) = \frac{1}{n_{\text{valid}}} \sum_{b=1}^{n_{\text{valid}}} \delta^{(b)}_{d,t_1} \delta^{(b)}_{d,t_2}.$$

It is computed in the same loop as the MSE, from the same draws, so the two are mutually consistent. Together they fill the diagonal (MSE) and off-diagonals (MCPE) of the per-domain prediction-error covariance matrix of $\left(\hat{\theta}^{\text{bench}}_{d,1} - \theta_{d,1}, \ldots, \hat{\theta}^{\text{bench}}_{d,T} - \theta_{d,T}\right)$. The MCPE is essential for valid inference on changes over time: the variance of an estimated change $\hat{\theta}^{\text{bench}}_{d,t_2} - \hat{\theta}^{\text{bench}}_{d,t_1}$ is $\widehat{MSE}_{d,t_2} + \widehat{MSE}_{d,t_1} - 2\,\widehat{MCPE}_d(t_1, t_2)$. When $T = 1$ the routine returns `mcpe_bench = NULL`.

## 6. How MFH1 and MFH3 differ

Only the random-effect data-generating process changes; the sampling-error draws, refit, re-benchmarking, accumulation, and division by $n_{\text{valid}}$ are identical. MFH1 replaces the AR(1) recursion with independent draws $u_{dt}^{(b)} \sim N(0, \hat{\sigma}_{u,t}^2)$ and refits with eblupMFH1. MFH3 keeps the AR(1) recursion but with time-varying innovation variance and the non-stationary period-1 initialization of Section 4.2, and refits with eblupMFH3. The model_type argument selects both the DGP and the refit function.

## 7. Function interface and returned object

bench_regional_mfh() accepts the EBLUP matrix, the direct-estimate matrix, domain and region identifiers, population sizes (Nd_vec, or the optional time-varying Nd_mat), an optional externally supplied regional_benchmark_mat, the fitted model_obj, the model_type, the formula/vardir/data needed to refit, and the bootstrap controls MSE, nB, seed, and max_attempts. It returns:

Returned components of bench_regional_mfh().

| Element | Dimension | Description |
| --- | --- | --- |
| eblup_bench | $D \times T$ | Benchmarked EBLUPs $\hat{\theta}_{dt}^{\text{bench}}$. |
| mse_bench | $D \times T$ | Bootstrap MSE of the benchmarked EBLUPs (NULL if MSE = FALSE). |
| mcpe_bench | $D \times \binom{T}{2}$ | Bootstrap MCPE of the benchmarked EBLUPs (NULL if $T < 2$). |
| lambda | $D \times T$ | Ratio adjustment factors $\lambda_{rt}$ broadcast to domain rows. |
| fails | scalar | Number of bootstrap refits that failed to converge. |

Both eblup_mat and direct_mat (and any optional Nd_mat or regional_benchmark_mat) carry domain row names and time column names, and the routine checks that these match domain_vec and the EBLUP column order before computing anything, so a misaligned input raises an error rather than producing silently wrong results (Section 9, point 6).

## 8. Practical notes

The bootstrap is the only source of uncertainty quantification for the benchmarked predictor: the analytic MSE returned by eblupMFH2 applies to the unbenchmarked EBLUP and is not propagated through the ratio adjustment. The bootstrap is *parametric* — random effects are simulated from the fitted $N(0, \hat{\sigma}_u^2)$ law (with the AR(1) recursion for MFH2 and MFH3) and sampling errors from $N_T(0, \Sigma_{e,d})$ — whereas the univariate bench_regional() uses a nonparametric residual-resampling bootstrap; the re-benchmarking and scoring logic is otherwise identical. Because every replicate refits the model, the routine is computationally heavy; the default $n_B = 200$ balances Monte Carlo error against runtime and can be raised for production runs. The seed (default 123) controls the whole bootstrap, so repeated calls with the same inputs are reproducible. The unbenchmarked counterpart, which omits the re-benchmarking step, is pbmcpeMFH2() in scripts/mcpe_functions.R and shares the same data-generating process, which is convenient for sanity checks.

## 9. Review comments and their implementation

The six points raised in review (parametric bootstrapping and benchmarking) are all addressed in the current code, as summarized below.

Status of review comments on bench_regional_mfh().

| # | Review comment | Status | Where in the code |
|---|---|---|---|
| 1 | Allow regional benchmarks to be supplied as a function input, not only computed internally. | Implemented | regional_benchmark_mat argument; benchmarks read from a file via .matrix_from_optional_regional_benchmark() and a regional_benchmark_path config. |
| 2 | Population size could vary across time, not just $N_d$. | Implemented | Optional Nd_mat ($D \times T$); the routine uses $N_{dt}$ per period and falls back to a constant $N_d$ when Nd_mat is NULL. |
| 3 | Dividing by nB undercounts failed replications and underestimates MSE/MCPE. | Implemented | The loop retries failures and divides the accumulator by n_valid (successful replicates); failures are counted in fails. |
| 4 | Stationary AR(1) initialization is not exact for the heteroskedastic MFH3. | Implemented | MFH2 keeps the stationary initialization; MFH3 initializes period 1 directly with its period-1 innovation variance ($u_{d,1} = a_{d,1}$), imposing no stationarity. |
| 5 | The benchmark and the EBLUP regional mean used inconsistent domain subsets. | Implemented | Both $B_{rt}$ and $\hat{\theta}_{rt}^{\text{EBLUP}}$ are computed over the same basis_rows (domains with finite direct, EBLUP, and weight). |
| 6 | The direct and EBLUP matrices must share domain and time ordering, or results are silently wrong. | Implemented | Both matrices are assigned matching domain/time names; bench_regional_mfh() checks row/column names against domain_vec and the EBLUP order, |

| | |
|---|---|
| | and the caller additionally verifies the outcome ordering. |

## 10. References


Benavent, R. and Morales, D. (2016). Multivariate Fay–Herriot models for small area estimation. *Computational Statistics and Data Analysis*, 94, 372–390.

Datta, G. S., Ghosh, M., Steorts, R., and Maples, J. (2011). Bayesian benchmarking with applications to small area estimation. *Test*, 20(3), 574–588.

Molina, I. and Romero, E. (2025). *Comparable Small Area Estimates over Time.* Working document.